USN										

"24CS310

JSS MAHAVIDYAPEETHA
JSS SCIENCE AND TECHNOLOGY UNIVERSITY, MYSURU

Department of Computer Science and Engineering

III Semester

Test I

Object Oriented Programming using C++

Duration: 1.15 Hr | Date: 10/11/2025 | Max. Marks: 30 Marks

NOTE: Question 1,4 & 5 are compulsory.
Answer any one between question 2 and question 3.

Q.No.	CO	CD	PI	QUESTION	MARKS
1a	CO1	L2	1.4.1	Explain the concept of inline functions with its advantages. Write a C++ program to calculate the sum of two numbers using inline function.	07
1b	CO1	L3	1.4.1	Write a C++ program to demonstrate function overloading by creating a function maxValue() that finds the largest value among: i) Two integers ii)Two floating-point numbers and iii) Three integers.	08
2a	CO2	L2	2.1.2	What is a constructor? Discuss the rules for defining constructors in C++. Write a suitable example to illustrate the use of a constructor.	05
2b	CO2	L3	2.1.2	Create a class *Sample* with two private integer data members. Use member function to read the data members. Write a C++ program to calculate and display the sum of these two members using a non-member function that is declared as a friend of the class.	05
				OR	
3a	CO2	L3	2.1.2	Write a C++ program to demonstrate the concept of reference variables by swapping two numbers using a function with reference parameters. Also explain how reference parameters differ from pointers or value parameters.	05
3b	CO2	L2	2.1.2	Explain default arguments with an example program.	05
4	CO1	L2	1.4.1	Explain any three key features of Object-Oriented Programming with suitable example.	03
5	CO2	L1	2.1.2	Differentiate between dynamic memory allocation and static memory allocation.	02

CO#	Course Outcomes	Highest Level of Cognitive Domain
CO1	Understand object-oriented programming concepts and be able to design solutions for a given problem.	L3
CO2	Able to reuse code with extensible Class types, user defined operators and function overloading.	L3
CO3	Achieve code reusability and extensibility by means of Inheritance and Polymorphism	L3
CO4	Incorporate file-handling functions and exceptions for a given application.	L3
CO5	Design C++ packages with templates for providing	L3

PI's	
1.4.1	Apply theory and principles of computer science and engineering to solve an engineering problem
2.1.2	Identify processes/modules/algorithms of a computer-based system and parameters to solve a problem

Blooms Taxonomy	
L1	Remembering
L2	Understanding
L3	Applying

— End —